★ Can all PSEs be identified from obs data?

- It is difficult without additional assumption.

In the case of 2 mediators

$$\begin{tikzcd} A \arrow[r] \arrow[rr, bend left] \arrow[rrr, bend right=40] & M_1 \arrow[r] \arrow[rr, bend right] & M_2 \arrow[r] & Y \end{tikzcd}$$

M_1 is obvious to be a intermediated confounder in the relationship among A, M_2, Y

$\Rightarrow$ We cannot separate $\begin{cases} A \rightarrow M_1 \rightarrow Y \\ A \rightarrow M_1 \rightarrow M_2 \rightarrow Y \end{cases}$

★ $Y(a, M_1(e_1), M_2(e_2, M_1(e_3)))$?

$$\begin{tikzcd} & & & \varepsilon_Y \arrow[d] \\ A \arrow[r] \arrow[rr, bend left] \arrow[rrr, bend left=40] & M_1 \arrow[r] \arrow[rr, bend right] & M_2 \arrow[r] & Y \\ \varepsilon_A \arrow[u] & \varepsilon_1 \arrow[u] & \varepsilon_2 \arrow[u] & \end{tikzcd}$$

$A = g_A(\varepsilon_A)$
$M_1 = g_1(A, \varepsilon_1)$
$M_2 = g_2(A, M_1, \varepsilon_2)$
$Y = g_Y(A, M_1, M_2, \varepsilon_Y)$
$\Downarrow$
Let $A = 1$
$M_1(1) = g_1(1, \varepsilon_1)$
$M_2(1) = g_2(1, M_1(1), \varepsilon_2)$
$Y(1) = g_Y(1, M_1(1), M_2(1, M_1(1)), \varepsilon_Y)$

$$\begin{aligned} TE &= Y(1) - Y(0) \\ &= Y(1, M_1(1), M_2(1, M_1(1))) \\ &\quad - Y(0, M_1(0), M_2(0, M_1(0))) \end{aligned}$$

— path $A \rightarrow Y$
······ path $A \rightarrow M_1 \rightarrow Y$
- - - - path $A \rightarrow M_2 \rightarrow Y$
-·-·- path $A \rightarrow M_1 \rightarrow M_2 \rightarrow Y$

Decomposition Strategies

1. Two-way
2. Parallel

3. Partial Forward
4. 〃 Backward
5. Complete

★ Two-way Decomposition

All mediators are treated as one multivariate mediator

$$\underset{\sim}{M} = (M_1, \dots, M_p)$$

TE is decomposed into $\begin{cases} \text{passing through } \underset{\sim}{M} \\ \text{not passing through } \underset{\sim}{M} \end{cases}$

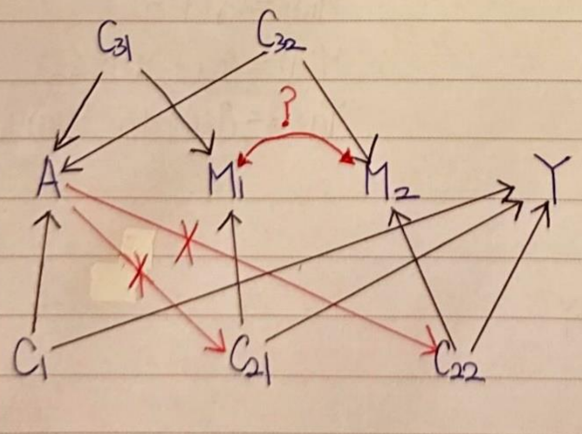
$\Rightarrow$

$$\begin{aligned} TE &= \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] \\ &= \underbrace{\left(\mathbb{E}[Y(1, \underset{\sim}{M}(1))] - \mathbb{E}[Y(1, \underset{\sim}{M}(0))]\right)}_{NIE} \\ &\quad + \underbrace{\left(\mathbb{E}[Y(1, \underset{\sim}{M}(0))] - \mathbb{E}[Y(0, \underset{\sim}{M}(0))]\right)}_{NDE} \end{aligned}$$

Define $\phi_{TW}(a, a^*) = \mathbb{E}[Y(a, \underset{\sim}{M}(a^*))]$

$$\begin{aligned} TE = \phi_{TW}(1,1) - \phi_{TW}(0,0) &= [\phi_{TW}(1,1) - \phi_{TW}(1,0)] \\ &+ [\phi_{TW}(1,0) - \phi_{TW}(0,0)] \end{aligned}$$

Identification of $\phi_{TW}(a, a^*)$



$Y(a, \underset{\sim}{m}) \perp\!\!\!\perp A \mid C = C_1$

$Y(a, \underset{\sim}{m}) \perp\!\!\!\perp \underset{\sim}{M} \mid A, C = (C_{21}, C_{22})$

$\underset{\sim}{M}(a) \perp\!\!\!\perp A \mid C = (C_{31}, C_{32})$

$Y(a, \underset{\sim}{m}) \perp\!\!\!\perp \underset{\sim}{M}(a^*) \mid C$

$$\phi_{TW}(a, a^*) = \sum_{c} \sum_{\underset{\sim}{m}} \mathbb{E}(Y \mid \underset{\sim}{M} = \underset{\sim}{m}, A = a, C = c)\, P(\underset{\sim}{M} = \underset{\sim}{m} \mid A = a^*, C = c)\, P(C = c)$$

Estimation

$$E(M_1|A) = \theta_0 + \theta_1 A$$
$$E(M_2|A) = \beta_0 + \beta_1 A$$
$$E(Y|M_2, M_1, A) = \alpha_0 + \alpha_1 M_1 + \alpha_2 M_2 + \alpha_3 A$$

$\Rightarrow$

$$\phi_{TN}(a, a^*) = \sum_{\underset{\sim}{m}} (\alpha_0 + \alpha_1 m_1 + \alpha_2 m_2 + \alpha_3 a) P(\underset{\sim}{M} = \underset{\sim}{m} \mid A = a^*)$$
$$= \alpha_0 + \alpha_3 a + \alpha_1(\theta_0 + \theta_1 a^*) + \alpha_2(\beta_0 + \beta_1 a^*)$$
$$= (\alpha_0 + \alpha_1\theta_0 + \alpha_2\beta_0) + \alpha_3 a + (\alpha_1\theta_1 + \alpha_2\beta_1) a^*$$

$$NIE_{TN} = \phi_{TN}(1,1) - \phi_{TN}(1,0) = \alpha_1\theta_1 + \alpha_2\beta_1 = [\theta_1 \ \beta_1] \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}$$
$$NDE_{TN} = \phi_{TN}(1,0) - \phi_{TN}(0,0) = \alpha_3$$

$$\begin{tikzcd} A \arrow[r, "(\theta_1, \beta_1)"] \arrow[rr, bend left, "\alpha_3"] & \underset{\sim}{M} = (M_1, M_2) \arrow[r, "(\alpha_1, \alpha_2)"] & Y \end{tikzcd}$$

☆ Parallel Decomposition

Consider non-ordered mediators $M_1, \ldots, M_p$, meaning that the causal relationship between mediators is absent.

$$TE = E[Y(1)] - E[Y(0)]$$
$$= E[Y(1, M_1(1), M_2(1))] - E[Y(0, M_1(0), M_2(0))]$$
$$= \Big[E[Y(1)] - E[Y(1, M_1(1), M_2(0))]\Big] \Leftrightarrow PSE_2 \qquad (PSE_{12} = 0)$$
$$+ \Big[E[Y(1, M_1(1), M_2(0))] - E[Y(1, M_1(0), M_2(0))]\Big] \Leftrightarrow PSE_1$$
$$+ E[Y(1, M_1(0), M_2(0))] - E[Y(0)] \Leftrightarrow PSE_0$$

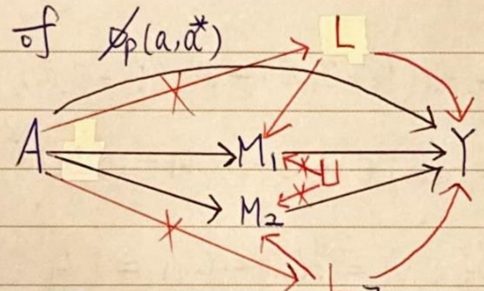
Define $\phi_p(a, e_1, e_2) = E[Y(a, M_1(e_1), M_2(e_2))]$

$$\Updownarrow$$

$$E[Y(a, M_1(e_1), M_2(e_2, M_1(e_1)))]$$

$$TE = \phi_p(1,1,1) - \phi_p(0,0,0)$$
$$= [\phi_p(1,1,1) - \phi_p(1,1,0)]$$
$$+ [\phi_p(1,1,0) - \phi_p(1,0,0)]$$
$$+ [\phi_p(1,0,0) - \phi_p(0,0,0)]$$
$$= PSE_2 + PSE_1 + PSE_0$$

Idenfication of $\phi_p(a, a^*)$



$$\phi_p(a, e_1, e_2) = \mathbb{E}[Y(a, M_1(e_1), M_2(e_1))]$$
$$= \sum_{m_1}\sum_{m_2} \mathbb{E}[Y(a, M_1(e_1), M_2(e_2)) \mid M_1(e_1)=m_1, M_2(e_2)=m_2] \times P(M_1(e_1)=m_1, M_2(e_2)=m_2)$$

$$\text{(Consistency)} = \sum_{m_1}\sum_{m_2} \mathbb{E}[Y(a, m_1, m_2) \mid M_1(e_1)=m_1, M_2(e_2)=m_2] \times P(M_1(e_1)=m_1, M_2(e_2)=m_2)$$

$$\begin{pmatrix} Y(a,m_1,m_2) \\ \perp\!\!\!\perp (M_1(e_1), M_2(e_2)) \end{pmatrix} = \sum_{m_1}\sum_{m_2} \mathbb{E}[Y(a,m_1,m_2)]\, P(M_1(e_1)=m_1, M_2(e_2)=m_2)$$

$$\left(M_1(e_1) \perp\!\!\!\perp M_2(e_2)\right) = \sum_{m_1}\sum_{m_2} \mathbb{E}[Y(a,m_1,m_2)]\, P(M_1(e_1)=m_1)\, P(M_2(e_2)=m_2)$$

$$\begin{pmatrix} Y(a,m_1,m_2) \perp\!\!\!\perp A \\ Y(a,m_1,m_2) \perp\!\!\!\perp M_1 \mid A \\ Y(a,m_1,m_2) \perp\!\!\!\perp M_2 \mid A, M_1 \end{pmatrix} = \sum_{m_1}\sum_{m_2} \mathbb{E}[Y(a,m_1,m_2) \mid A=a, M_1=m_1, M_2=m_2]\, P(M_1(e_1)=m_1)\, P(M_2(e_2)=m_2)$$

$$\text{(Consistency)} = \sum_{m_1}\sum_{m_2} \mathbb{E}(Y \mid A=a, M_1=m_1, M_2=m_2)\, P(M_1(e_1)=m_1)\, P(M_2(e_2)=m_2)$$

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$$\begin{pmatrix} M_1(e_1) \perp\!\!\!\perp A \\ M_2(e_2) \perp\!\!\!\perp A \end{pmatrix} = \sum_{m_1}\sum_{m_2} E(Y \mid M_1=m_1, M_2=m_2, A=a)\, P(M_1(e_1)=m_1 \mid A=e_1) \times P(M_2(e_2)=m_2 \mid A=e_2)$$

$$(\text{consistency}) = \sum_{m_1}\sum_{m_2} E(Y \mid M_1=m_1, M_2=m_2, A=a)\, P(M_2=m_2 \mid A=e_2) \times P(M_1=m_1 \mid A=e_1)$$

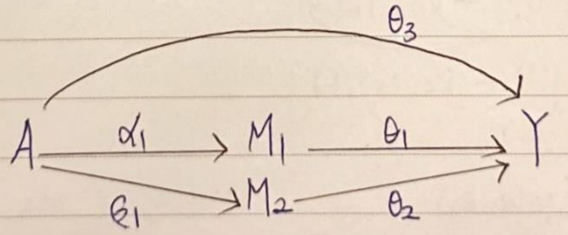
Estimation

$$E(Y \mid M_2, M_1, A) = \theta_0 + \theta_1 M_1 + \theta_2 M_2 + \theta_3 A$$
$$E(M_1 \mid A) = \alpha_0 + \alpha_1 A$$
$$E(M_2 \mid A) = \beta_0 + \beta_1 A$$

$\Rightarrow$

$$\phi_p(a, e_1, e_2) = \sum_{m_1}\sum_{m_2} (\theta_0 + \theta_1 m_1 + \theta_2 m_2 + \theta_3 a)\, P(M_2=m_2 \mid A=e_2) \times P(M_1=m_1 \mid A=e_1)$$
$$= \theta_0 + \theta_3 a + \theta_1(\alpha_0 + \alpha_1 e_1) + \theta_2(\beta_0 + \beta_1 e_2)$$
$$= (\theta_0 + \theta_1\alpha_0 + \theta_2\beta_0) + \theta_3 a + \theta_1\alpha_1 e_1 + \theta_2\beta_1 e_2$$

$$PSE_0 = \phi_p(1,0,0) - \phi_p(0,0,0) = \theta_3$$
$$PSE_1 = \phi_p(1,1,0) - \phi_p(1,0,0) = \theta_1\alpha_1$$
$$PSE_2 = \phi_p(1,1,1) - \phi_p(1,1,0) = \theta_2\beta_1$$



★ Partial Forward (PF)

TE, $\underline{M}=(M_1,M_2)$:
- $PSE_0 \Leftrightarrow NDE_{12}$
- $PSE_1 + PSE_2 + PSE_{12} \Leftrightarrow NIE_{12}$

TE, $\underline{M} = M_1$:
- $PSE_0 + PSE_2 \Leftrightarrow NDE_1$
- $PSE_1 + PSE_{12} \Leftrightarrow NIE_1$

$PSE_0 = NDE_{12}$

PSE_1 (x)

$PSE_2 = NDE_1 - NDE_{12}$
$\quad = NIE_{12} - NIE_1$

PSE_{12} (x)

Double A

$Y(1, M_1(0), M_2(0, M_1(0)))$ $\quad$ $Y(0, M_1(0), M_2(0, M_1(0)))$

$$
\begin{aligned}
TE = &\Big(\mathbb{E}[Y(1, M_1(0), M_2(0))] - \mathbb{E}[Y(0)]\Big) \iff PSE_0 \\
&+\Big(\mathbb{E}[Y(1, M_1(0))] - \mathbb{E}[Y(1, M_1(0), M_2(0))]\Big) \iff PSE_2 \\
&+\Big(\mathbb{E}[Y(1)] - \mathbb{E}[Y(1, M_1(0))]\Big) \iff PSE_1 + PSE_{12}
\end{aligned}
$$

$Y(1, M_1(0), M_2(0, M_1(0)))$

$Y(1, M_1(1), M_2(1, M_1(1)))$ $\quad$ $Y(1, M_1(0), M_2(1, M_1(0)))$

Define $\cancel{\phi}_c(a, e_1, e_2, e_3) = \mathbb{E}[Y(a, M_1(e_1), M_2(e_2, M_1(e_3)))]$

$$
\begin{aligned}
\Rightarrow \quad TE = \;& \cancel{\phi}_c(1,0,0,0) - \cancel{\phi}_c(0,0,0,0) \\
&+ \cancel{\phi}_c(1,0,1,0) - \cancel{\phi}_c(1,0,0,0) \\
&+ \cancel{\phi}_c(1,1,1,1) - \cancel{\phi}_c(1,0,1,0)
\end{aligned}
$$

$$\cancel{\phi}_s(a, e_1, e_2) = \mathbb{E}[Y(a, M_1(e_1), M_2(e_2, M_1(e_1)))] = \cancel{\phi}_c = (a, e_1, e_2, e_1)$$

$$
\begin{aligned}
\Rightarrow \quad TE = \;& \cancel{\phi}_s(1,0,0) - \cancel{\phi}_s(0,0,0) \\
&+ \cancel{\phi}_s(1,0,1) - \cancel{\phi}_s(1,0,0) \\
&+ \cancel{\phi}_s(1,1,1) - \cancel{\phi}_s(1,0,1)
\end{aligned}
$$

The identfication of $\cancel{\phi}_s(a, e_1, e_2)$

$$
\begin{aligned}
\cancel{\phi}_s(a, e_1, e_2) &= \mathbb{E}[Y(a, M_1(e_1), M_2(e_2, M_1(e_1)))] \\
&= \sum_{m_1}\sum_{m_2} \mathbb{E}[Y(a, M_1(e_1), M_2(e_2, M_1(e_1))) \mid M_1(e_1) = m_1, M_2(e_2, M_1(e_1)) = m_2] \\
&\qquad \times P(M_1(e_1) = m_1, M_2(e_2, M_1(e_1)) = m_2) \\
\text{(Consistency)} &= \sum_{m_1}\sum_{m_2} \mathbb{E}[Y(a, m_1, m_2) \mid M_1(e_1) = m_1, M_2(e_2, M_1(e_1)) = m_2] \\
&\qquad \times P(M_2(e_2, M_1(e_1)) = m_2 \mid M_1(e_1) = m_1) \\
&\qquad \times P(M_1(e_1) = m_1)
\end{aligned}
$$